

CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Innovative and results-driven Machine Learning Engineering Lead with over 9 years of experience in software engineering, machine learning system architecture, and full lifecycle project execution. Proven track record of leading cross-functional teams to deploy scalable ML solutions in mission-critical environments. Adept at building MLOps pipelines, deploying AI/ML workloads in cloud environments (GCP, AWS), and applying advanced deep learning techniques using PyTorch and TensorFlow. Skilled in balancing technical vision, hands-on development, and strategic execution.

TECHNICAL SKILLS

- **Languages:** Python, Go, Rust, Java, TypeScript, R, Bash
- **ML Frameworks:** PyTorch, TensorFlow, TensorRT, scikit-learn, OpenCV, HuggingFace, spaCy
- **MLOps & Lifecycle:** ClearML, MLflow, Kubeflow, CI/CD, GitLab CI, Jenkins, Airflow
- **Data & ETL:** Pandas, NumPy, SQL, Apache Spark, Hadoop, BigQuery, Kafka, Flink
- **Cloud:** Google Cloud Platform (Vertex AI, GKE, BigQuery), AWS (S3, SageMaker), Azure
- **DevOps:** Docker, Kubernetes, Helm, Terraform, Ansible, Prometheus, Grafana
- **Tools:** Git, Jupyter, VS Code, IntelliJ, Notion, Jira, Confluence, Postman
- **Languages:** English (C2), Portuguese (C2), Romanian (native), Estonian (intermediate)

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Led a team of 5 in developing scalable machine learning inference services for maritime surveillance systems using TensorFlow Lite and PyTorch Mobile for edge deployments.
- Designed and implemented ClearML-based lifecycle pipelines for automating model experimentation, monitoring, versioning, and deployment, reducing ML deployment cycle time by 40%.
- Architected and deployed containerized ML microservices on GCP using GKE and Helm with integrated CI/CD via GitLab, achieving 99.95% system uptime.
- Developed anomaly detection models for critical infrastructure monitoring using unsupervised learning, improving fault detection rates by over 60%.
- Integrated telemetry streaming (Kafka + Spark) with real-time scoring models, enabling continuous risk profiling across multiple data channels.

- Collaborated with DevOps and data engineering teams to implement IaC solutions for reproducible ML workflows using Terraform and Kubernetes.
- Co-authored internal technical guidelines on MLOps adoption and edge AI performance tuning, adopted across multiple engineering teams.
- Delivered knowledge-sharing sessions and mentored 4 junior ML engineers in architecture patterns and ML deployment strategies.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Designed supervised learning models using scikit-learn and XGBoost to predict consumer behavior for digital marketing campaigns, boosting ROI by 28%.
- Integrated ML models into marketing automation pipelines using Flask REST APIs deployed in Docker containers across Azure Kubernetes Service.
- Led initiative to migrate legacy batch inference system to a stream-processing architecture using Kafka and Apache Flink, reducing latency by 75%.
- Collaborated with analysts and marketers to design feature stores using Spark-based ETL flows that generated 2TB+ weekly training data.
- Built telemetry dashboards using Grafana and Prometheus to monitor model inference metrics, data drift, and system health KPIs.
- Implemented A/B testing platform to validate model improvements with robust statistical tracking and automated rollback support.
- Designed and maintained reusable CI/CD pipelines for deploying ML APIs via GitLab CI and Docker Compose.
- Supported cross-functional teams by writing detailed technical documentation for model APIs, hyperparameter tuning protocols, and data validation schemas.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Developed and maintained machine learning-enabled SaaS modules in Python and Node.js, with a focus on document classification and image recognition.
- Contributed to a real-time vehicle tracking system by implementing object detection pipelines using YOLOv3 and OpenCV.
- Deployed training pipelines on AWS using EC2 spot instances and S3-backed data lake for multi-stage retraining workflows.
- Managed CI/CD automation with Jenkins and Docker, integrating automated model accuracy tests and cloud deployment hooks.
- Conducted code reviews and introduced PEP8, Black, and SonarQube standards across development teams.
- Contributed to GDPR-compliance by developing an explainable AI module that surfaces model decisions via SHAP and LIME techniques.
- Refactored large legacy systems to modular Python microservices, cutting average API response time by 42%.
- Participated in bi-weekly architecture sprints focused on transitioning systems to cloud-native services.

Software Developer
Decima Digital
November 2015 – April 2016
Tallinn, Estonia

- Developed RESTful backend services for ML-powered recommendation systems using Flask, PostgreSQL, and Redis.
- Designed Python-based batch data pipelines for product classification using NLP and TF-IDF vectorization.
- Worked on internal ML model versioning utility prior to adoption of modern platforms like ClearML.
- Contributed to the migration from local training setups to GCP infrastructure using preemptible GPU nodes.
- Evaluated and optimized logistic regression and SVM models for customer segmentation on clickstream data.
- Created CLI-based model retraining interface for use by non-technical analysts.
- Integrated daily auto-retraining jobs using cron and shell scripting into Jenkins workflow.
- Supported the implementation of model evaluation frameworks with support for F1-score, AUC, and confusion matrices.

Software Developer Intern
Borna Technology
September 2014 – August 2015
Tallinn, Estonia (On-Site)

- Assisted in developing machine learning proof-of-concepts including classification models using scikit-learn for fraud detection.
- Built internal tool for automated data cleaning and preprocessing for structured datasets using Pandas and NumPy.
- Contributed to ETL flows feeding a credit scoring model by developing reusable transformation functions.
- Wrote Python scripts for data ingestion from APIs and third-party datasets into PostgreSQL.
- Built visualizations in Matplotlib and Seaborn for reporting model insights to non-technical stakeholders.
- Participated in weekly stand-ups and sprint planning with the full engineering team, reporting technical blockers.
- Documented the intern onboarding process, including best practices for ML development and testing.
- Conducted performance benchmarking of various classification algorithms for model selection.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- ❖ TensorFlow Developer Certificate – TensorFlow by Google (2023)
- ❖ MLOps Specialization – DeepLearning.AI & AWS (2023)

- ❖ Google Cloud Professional Machine Learning Engineer – GCP (2024)
 - ❖ Data Science and Machine Learning Bootcamp – Udemy (2022)
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PROJECTS

- ✓ **Maritime AI Vision System**
Led architecture and deployment of computer vision models for ship detection and classification in satellite imagery. Combined CNNs (YOLOv5) with geospatial data and deployed models on edge devices with TensorRT.
- ✓ **AutoML Model Factory with ClearML**
Built a fully automated training pipeline using ClearML, GitLab CI, and GCP. Implemented model orchestration, parallel hyperparameter sweeps, performance dashboards, and alerting workflows.
- ✓ **Real-time ML Inference in Edge Drones**
Developed Python-based inference engine for embedded systems running lightweight object tracking models. Integrated with on-board telemetry and optimized for latency using ONNX.
- ✓ **Multilingual NLP Service with FastAPI**
Created language detection and sentiment analysis microservices supporting 10+ languages using spaCy, FastText, and FastAPI. Deployed with Docker and autoscaled on Kubernetes.